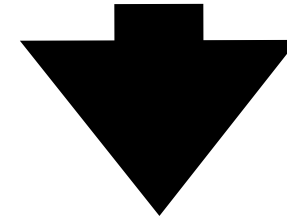
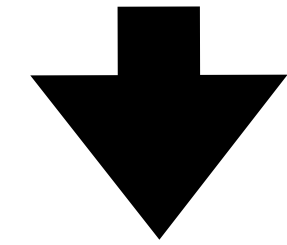


RECAP

Equivalence principle

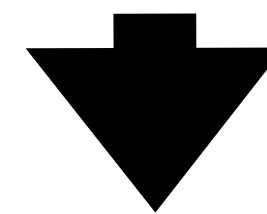


Gravity is geometry



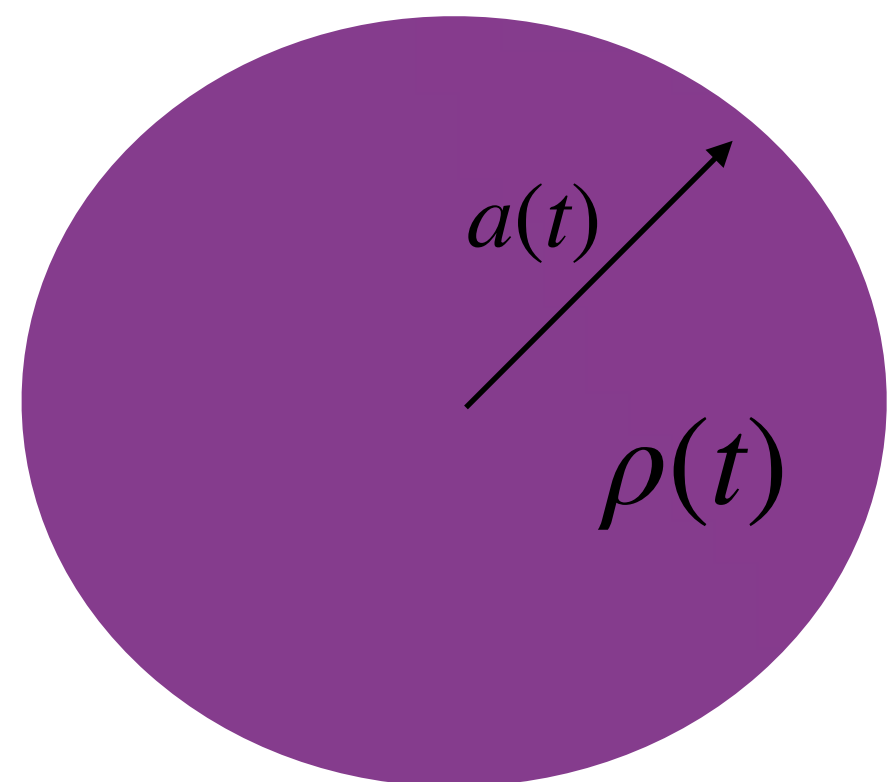
Spacetime curvature $\leftarrow R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu} \rightarrow$ Matter/energy content

Homogeneous and isotropic $g_{\mu\nu}$ and $T_{\mu\nu}$ Background cosmology



$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho$$
$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p)$$

Friedmann equations

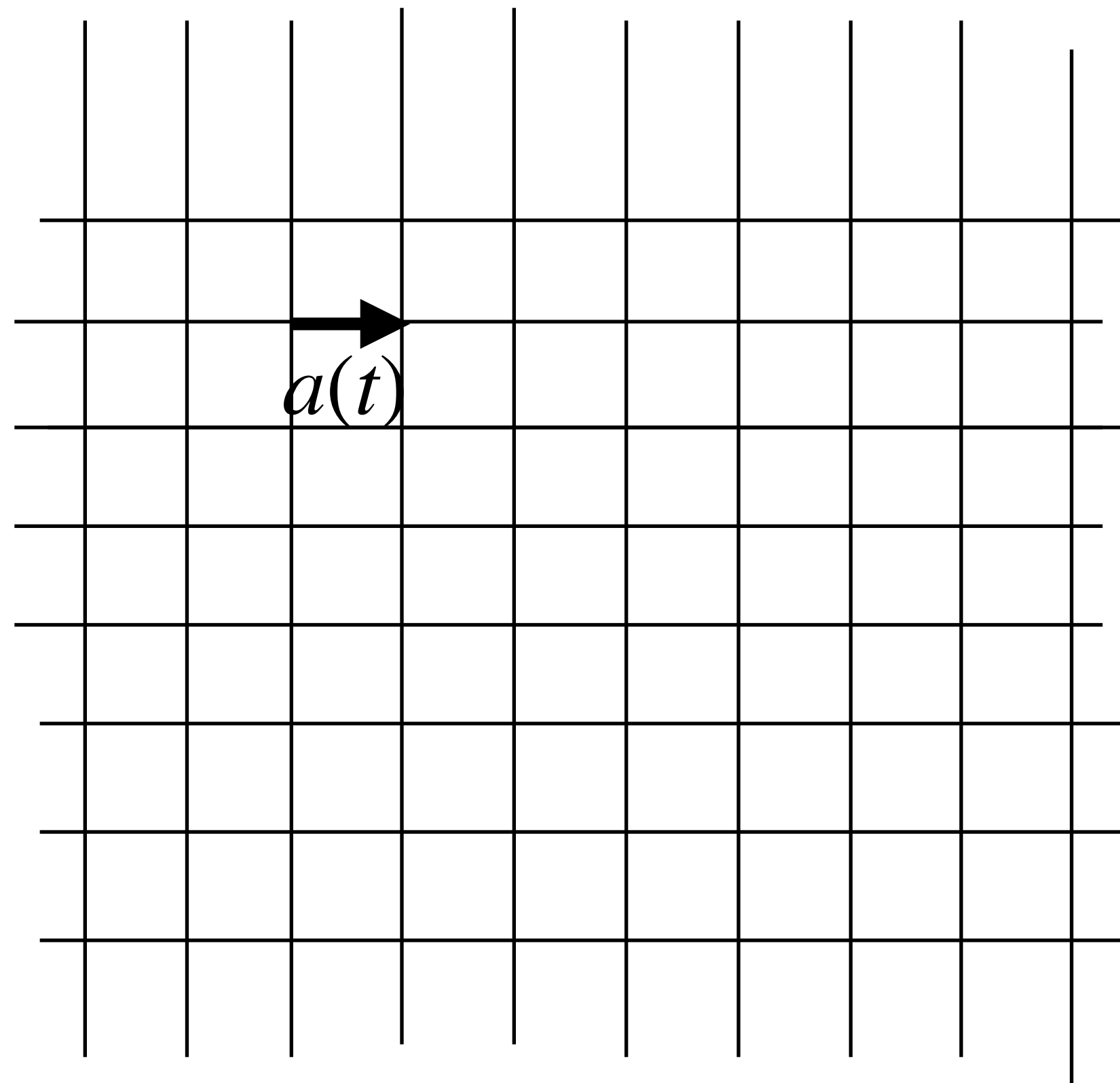
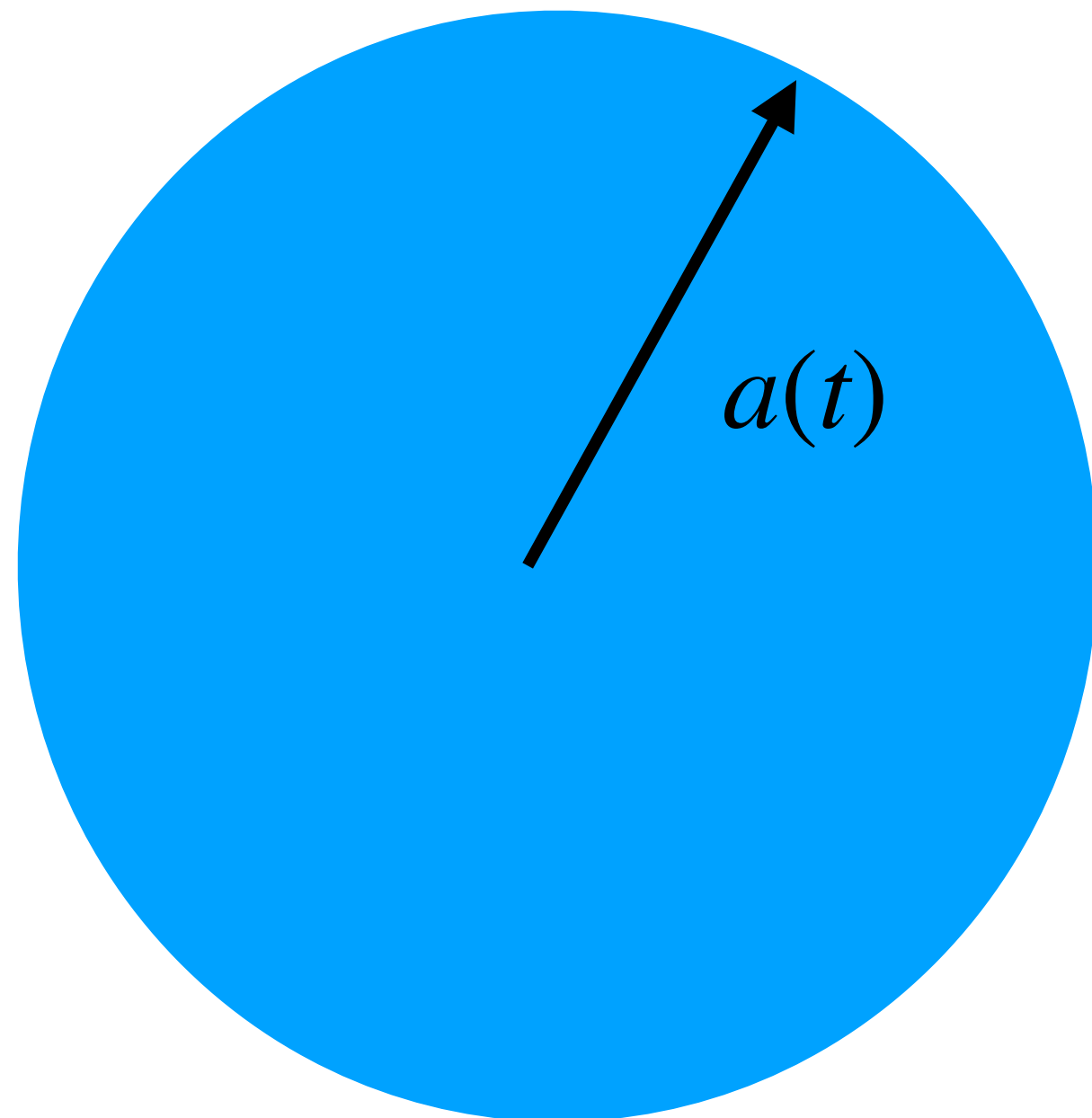


THE SCALE FACTOR

$$ds^2 = -dt^2 + a^2(t) \left[dr^2 + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right] \quad \text{Flat Friedmann-Lemaître-Robertson-Walker metric (FLRW)}$$

Or

$$ds^2 = -dt^2 + a^2(t) [dx^2 + dy^2 + dz^2] \quad \text{In cartesian coordinates}$$



We can define:

$$\dot{a} \equiv \frac{da}{dt} = \lim_{\Delta t \rightarrow 0} \frac{a(t + \Delta t) - a(t)}{\Delta t}$$

(The velocity with which the scale factor grows)

$$H(t) = \frac{1}{a} \frac{da}{dt} = \frac{\dot{a}}{a}$$

The Hubble factor:
The speed at which the Universe expands (or contracts)

THE SCALE FACTOR

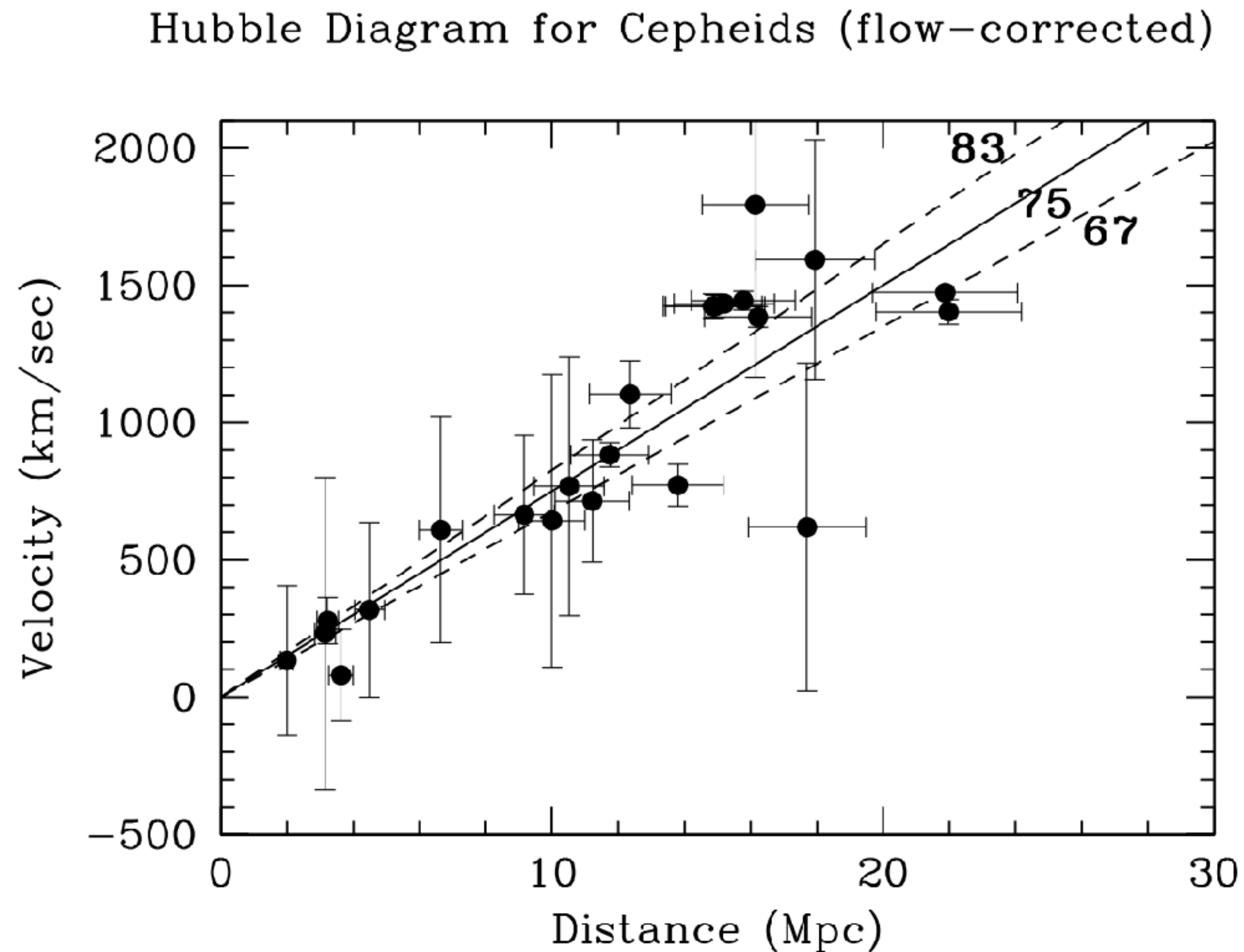
$$\left(\frac{1}{a} \frac{da}{dt}\right)^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \rho$$

$$\frac{da}{dt} = \sqrt{\frac{8\pi G}{3} \rho(a) a^2}$$

Give me what is your $\rho(a)$ and I give you how the Universe expands or contracts

GR + Homogeneous and isotropic Universe $\Rightarrow$ The prediction that the Universe is dynamical

THE EXPANDING UNIVERSE – HUBBLE'S LAW (1929)



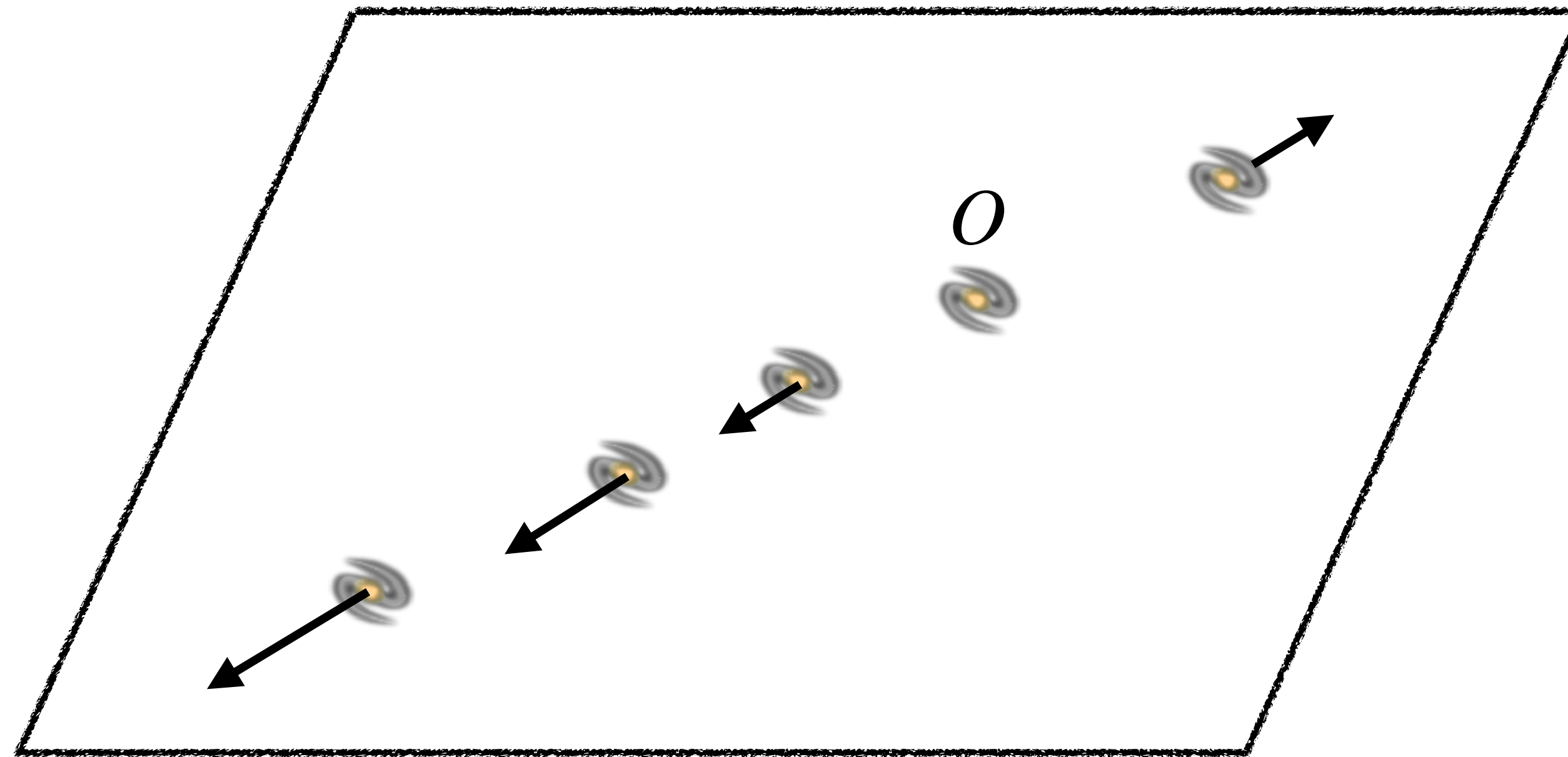
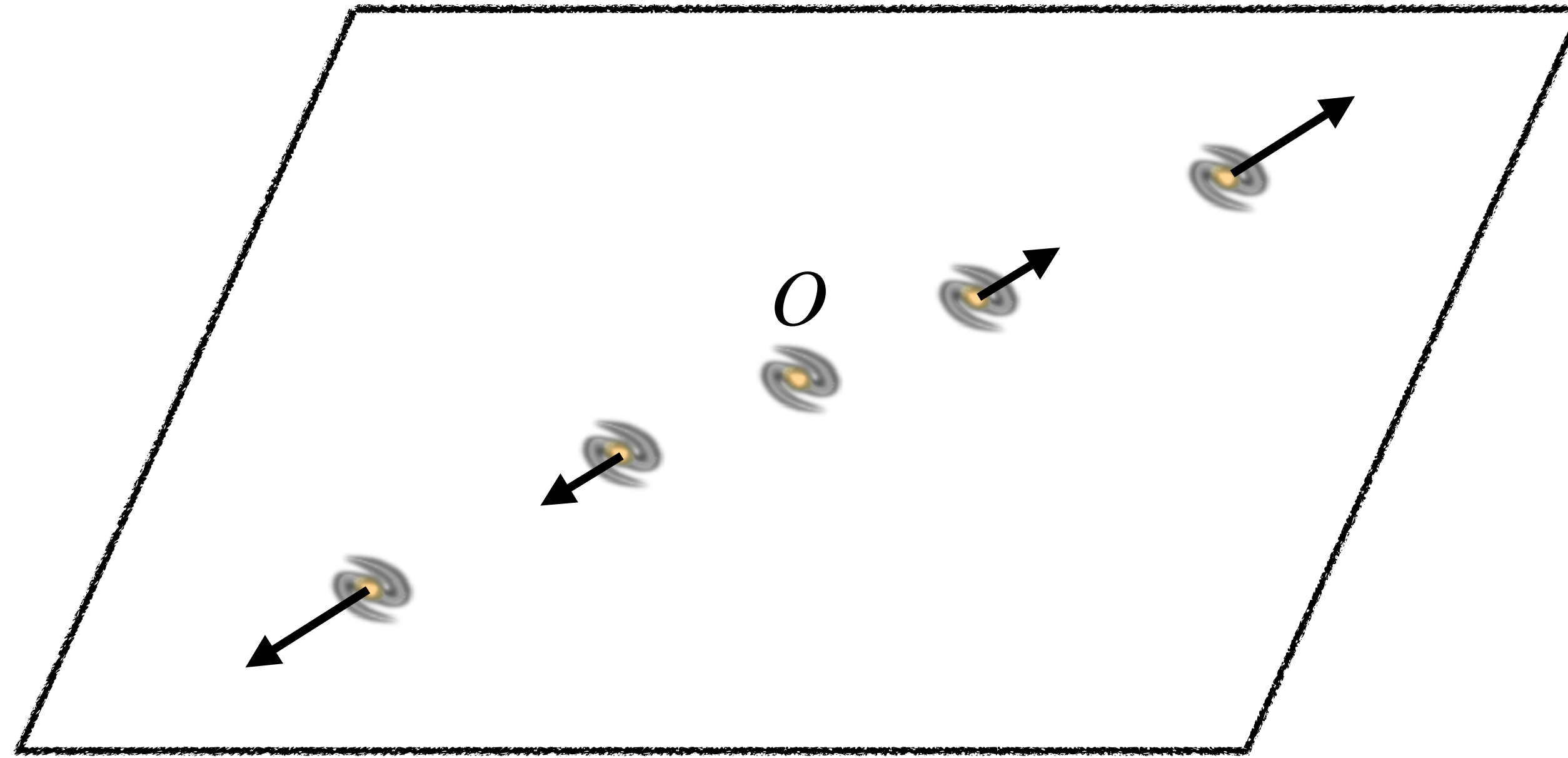
$$\frac{\lambda_{obs}}{\lambda_{em}} - 1 = \frac{a_0}{a(t)} - 1 = \frac{1}{a} - 1 = z$$

Scale factor - redshift relation:

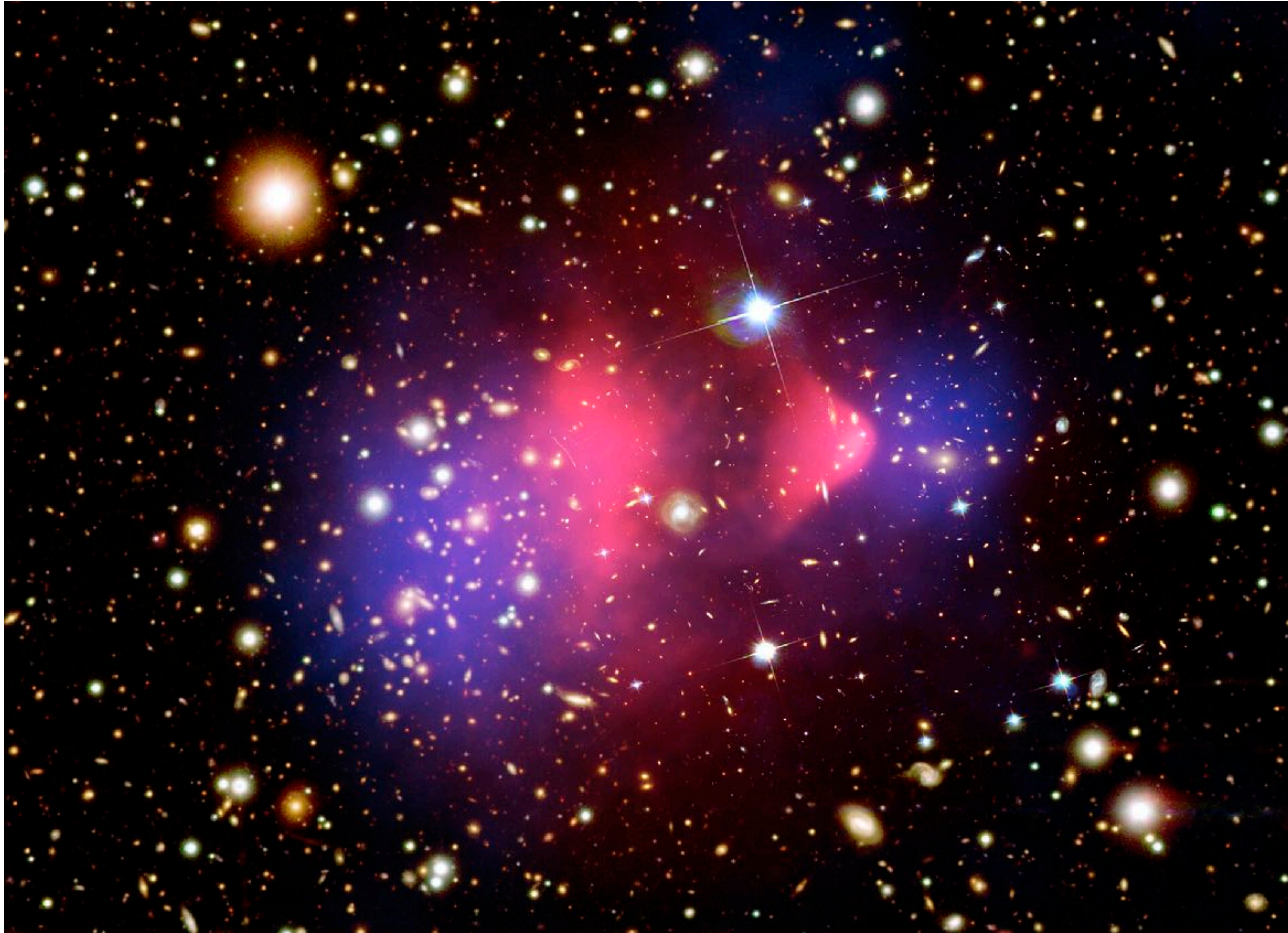
$$\frac{1}{a} = 1 + z$$

$$v = H_0 d \quad H_0 \simeq 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

THE EXPANDING UNIVERSE – HUBBLE'S LAW (1929)

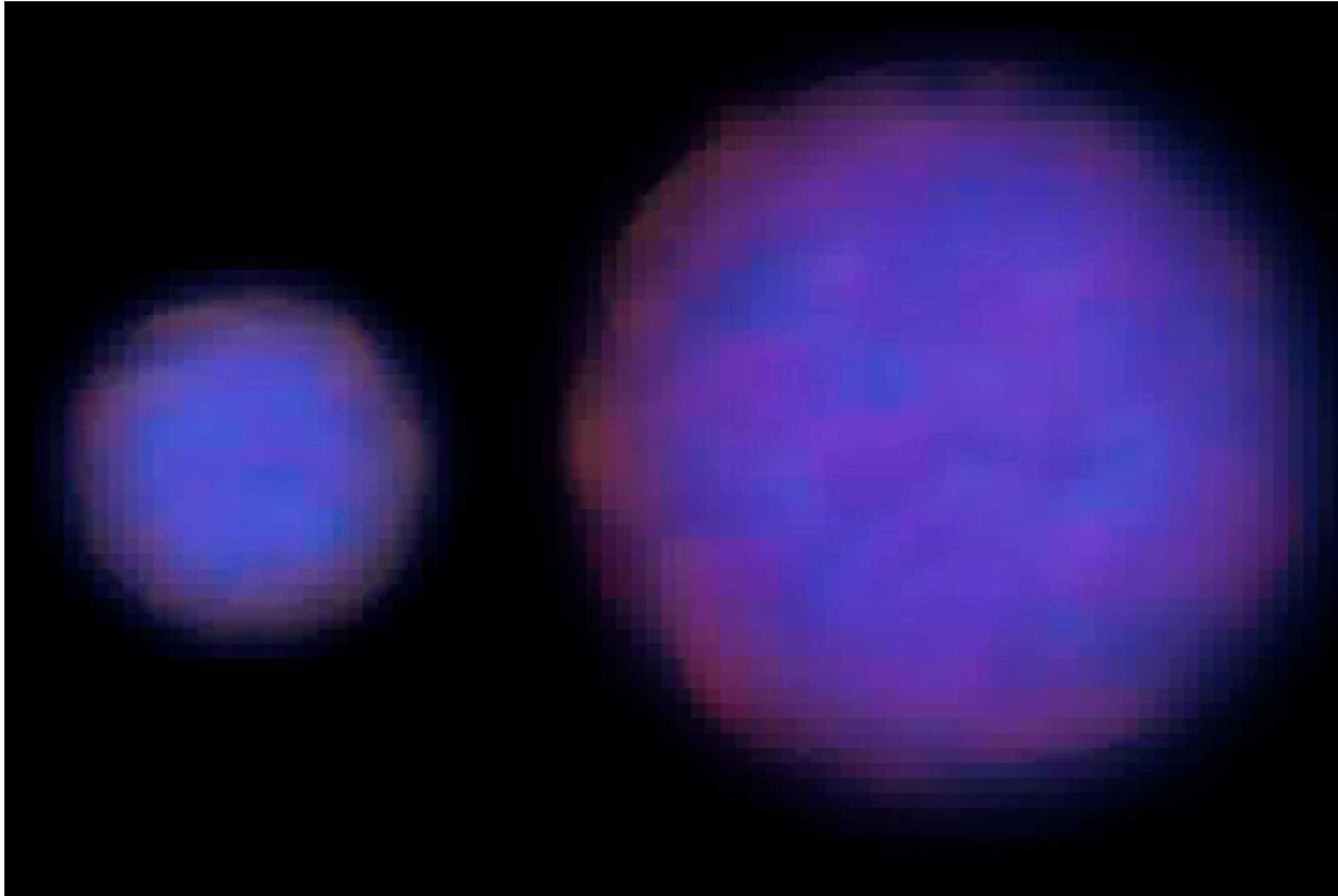


WHAT IS THE UNIVERSE MADE OF – DARK MATTER (THE BULLET CLUSTER)



Credit: ESA

WHAT IS THE UNIVERSE MADE OF – DARK MATTER (THE BULLET CLUSTER)



Credit: <https://www.youtube.com/watch?v=eC5Lwjsgl4I>

WHAT IS THE UNIVERSE MADE OF – DARK MATTER



https://www.astro.umd.edu/~diemer/erebos/web/viz/movies/movie8_br05.0_fps35.mp4

WHAT IS THE UNIVERSE MADE OF – DARK MATTER (CDM)

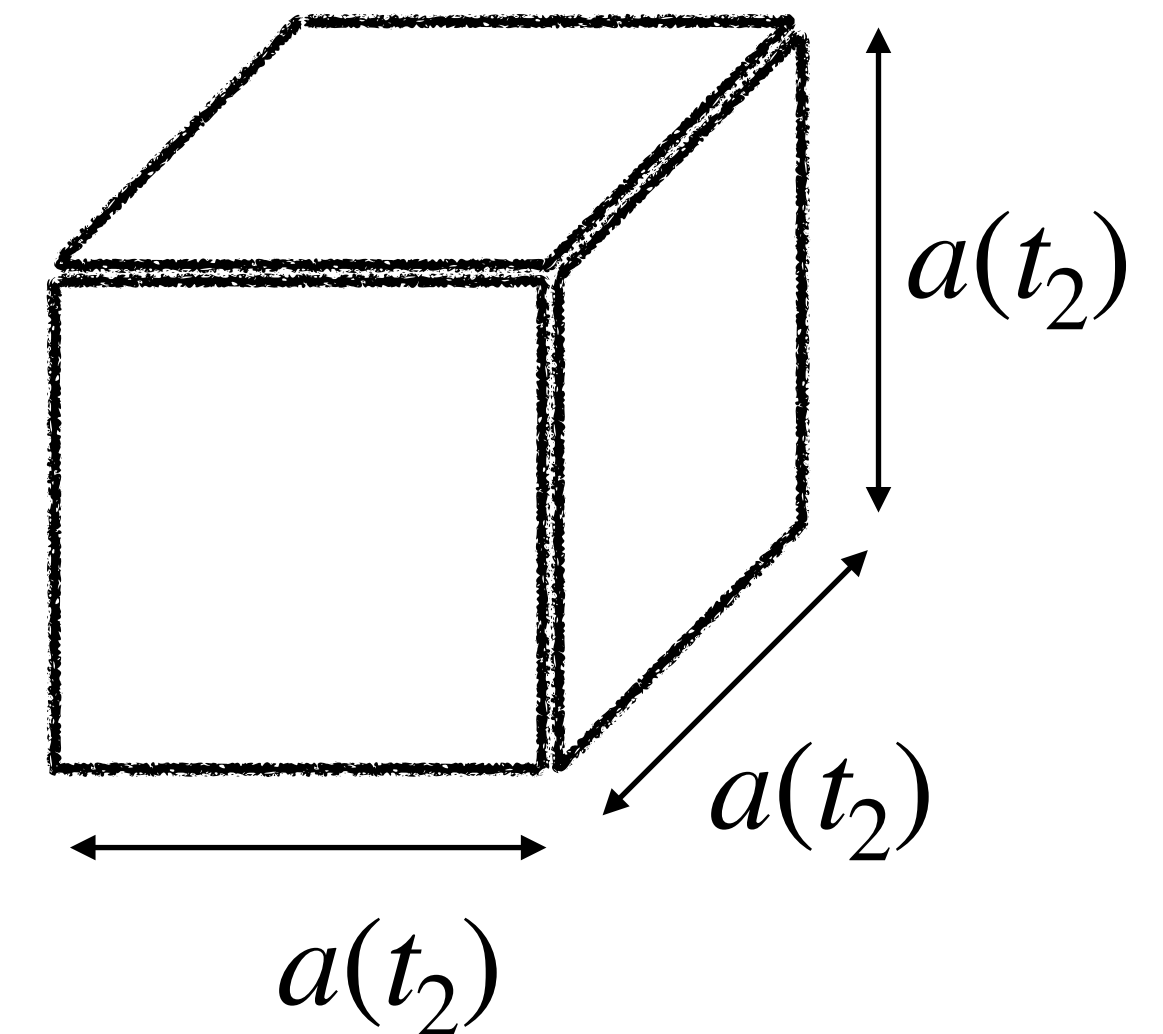
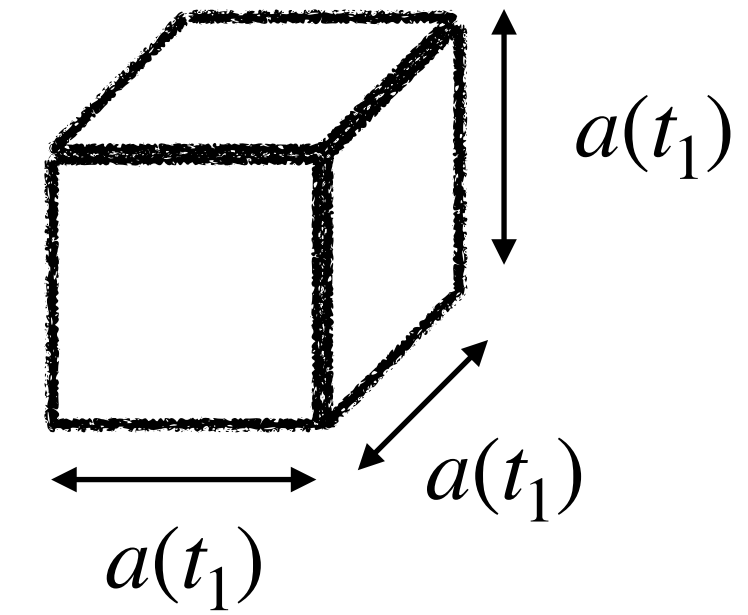
How does dark matter behave?

$$\rho = mn = m \frac{N}{V} \propto \frac{1}{a^3}$$

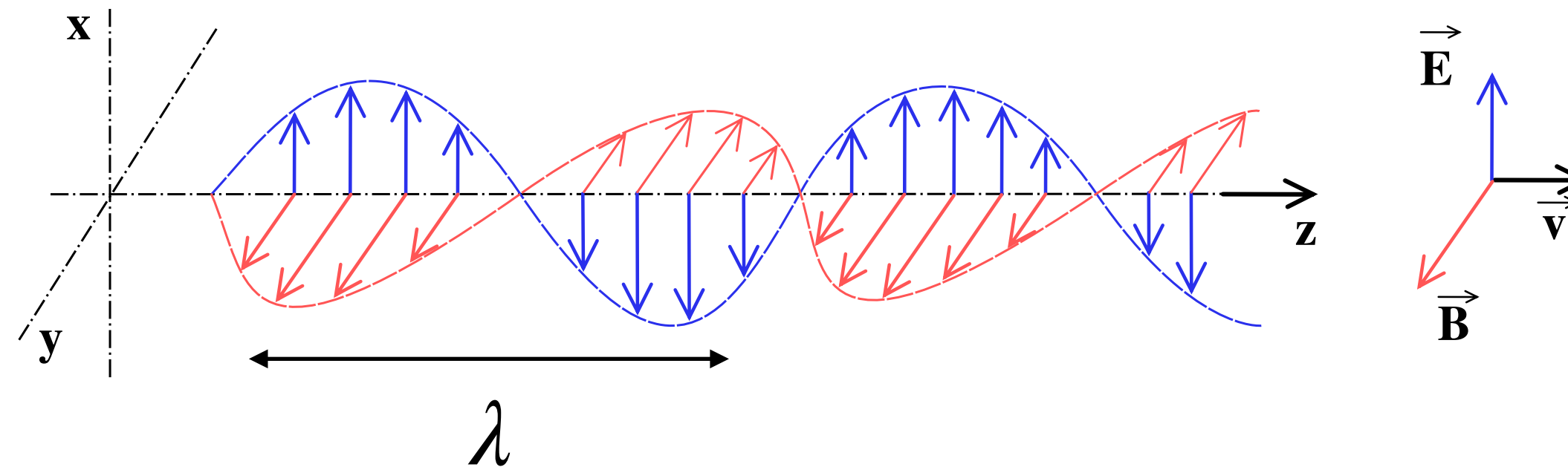
We can solve the Friedmann eq. for a DM-only Universe:

$$a(t) \propto t^{2/3}$$

It is cold ($v \ll c$) and pressureless ($p = 0$)

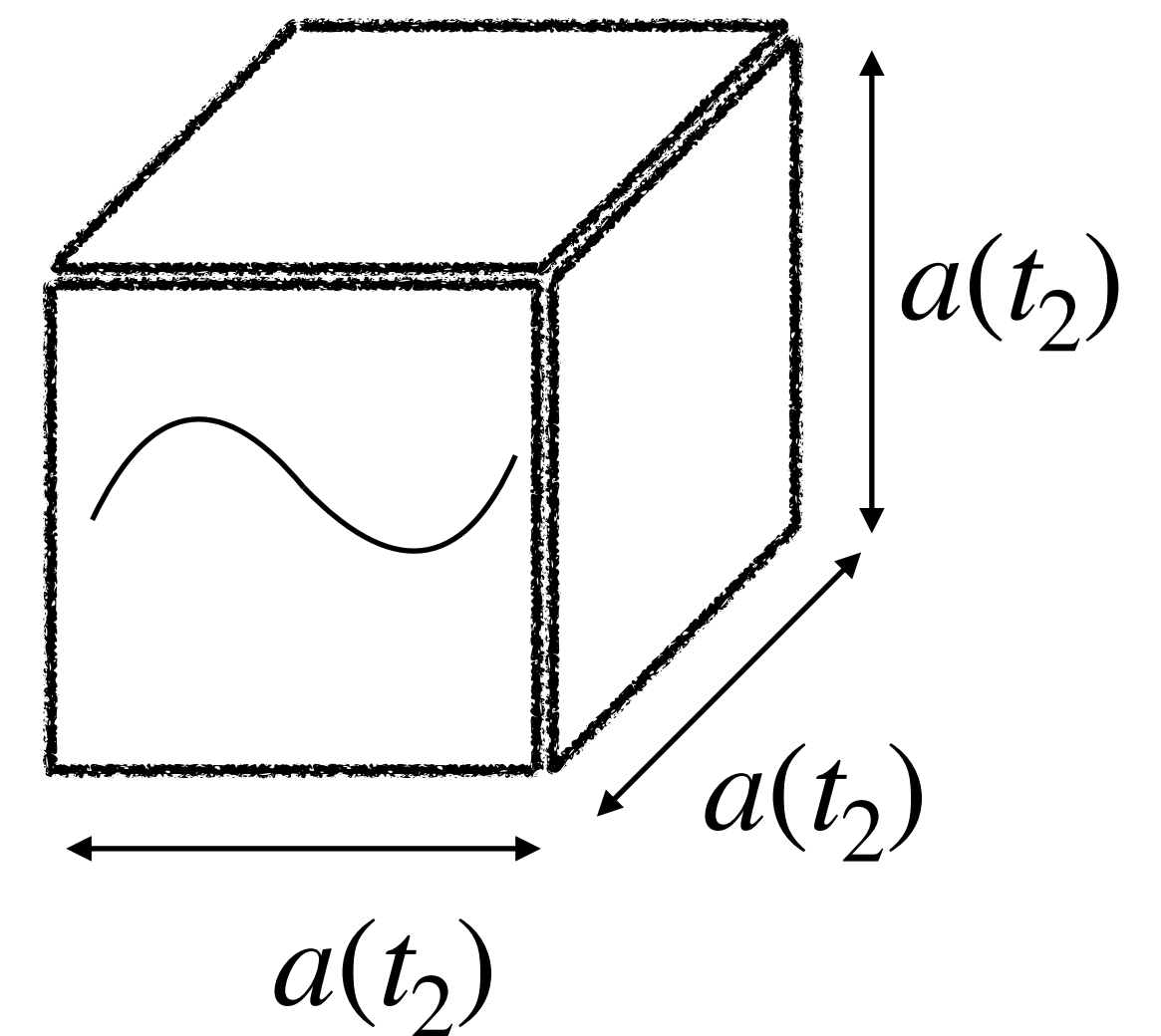
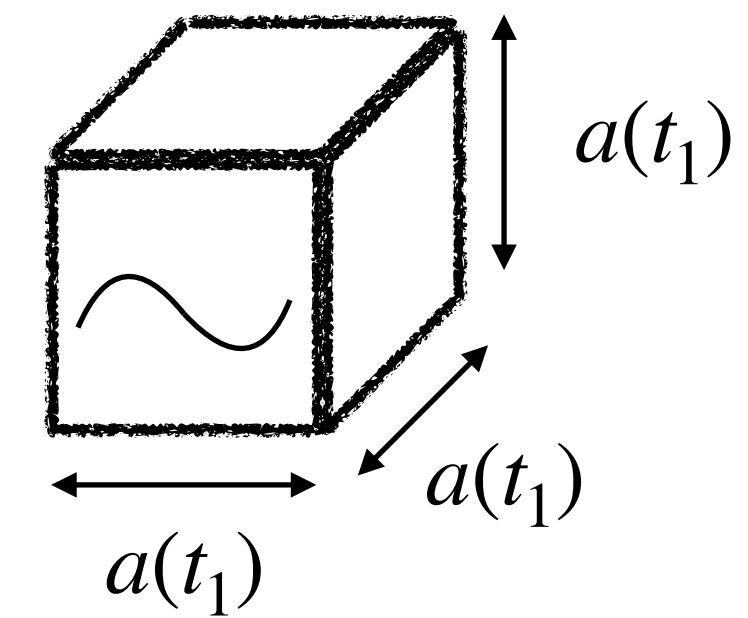
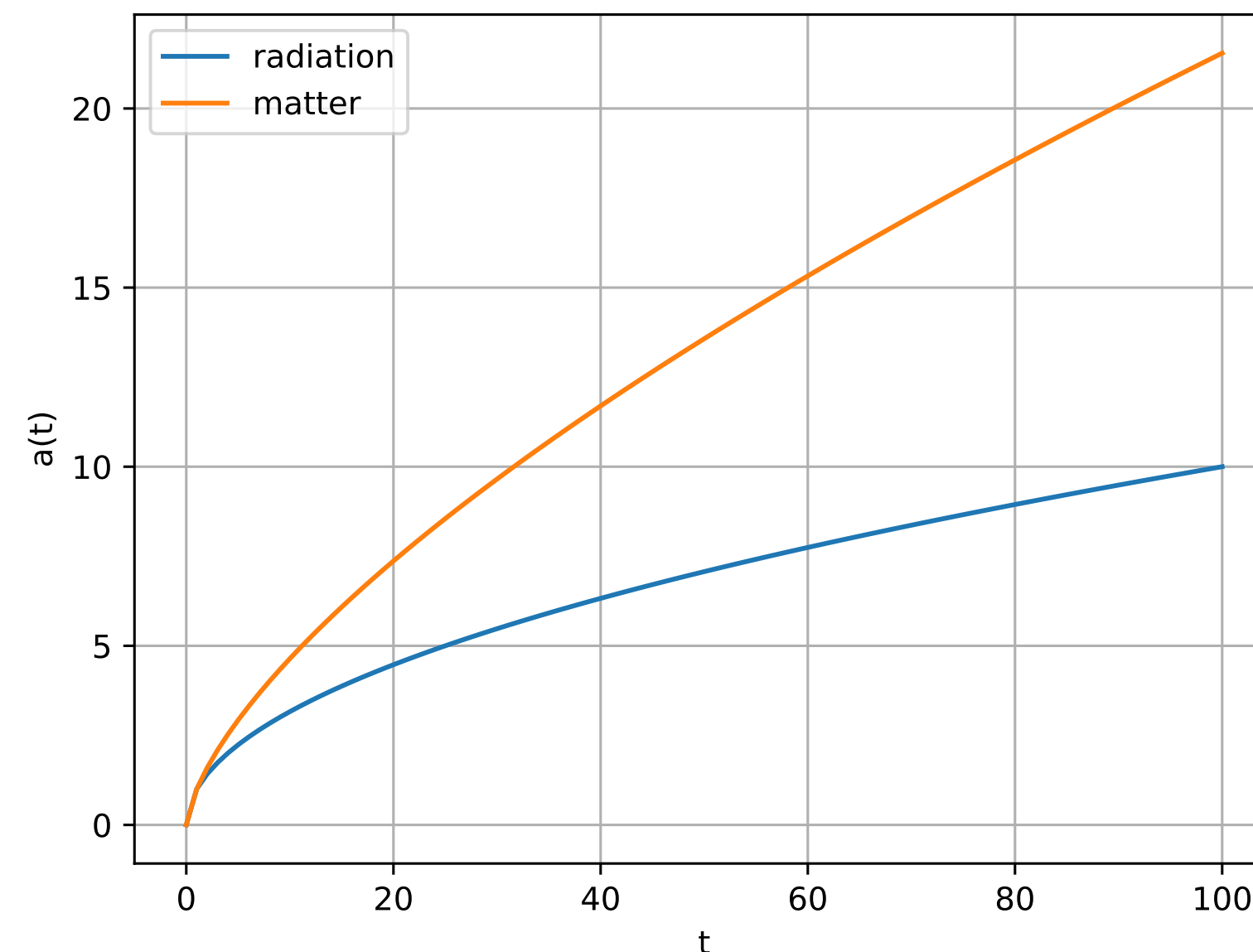


WHAT IS THE UNIVERSE MADE OF – RADIATION



Every particle with $v = c$, or $m = 0$

$$\rho_r = \frac{h}{\lambda} n = \frac{hN}{\lambda V} \propto a^{-4}$$



WHAT IS THE UNIVERSE MADE OF – DARK ENERGY

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \boxed{\Lambda g_{\mu\nu}} = 8\pi GT_{\mu\nu}$$

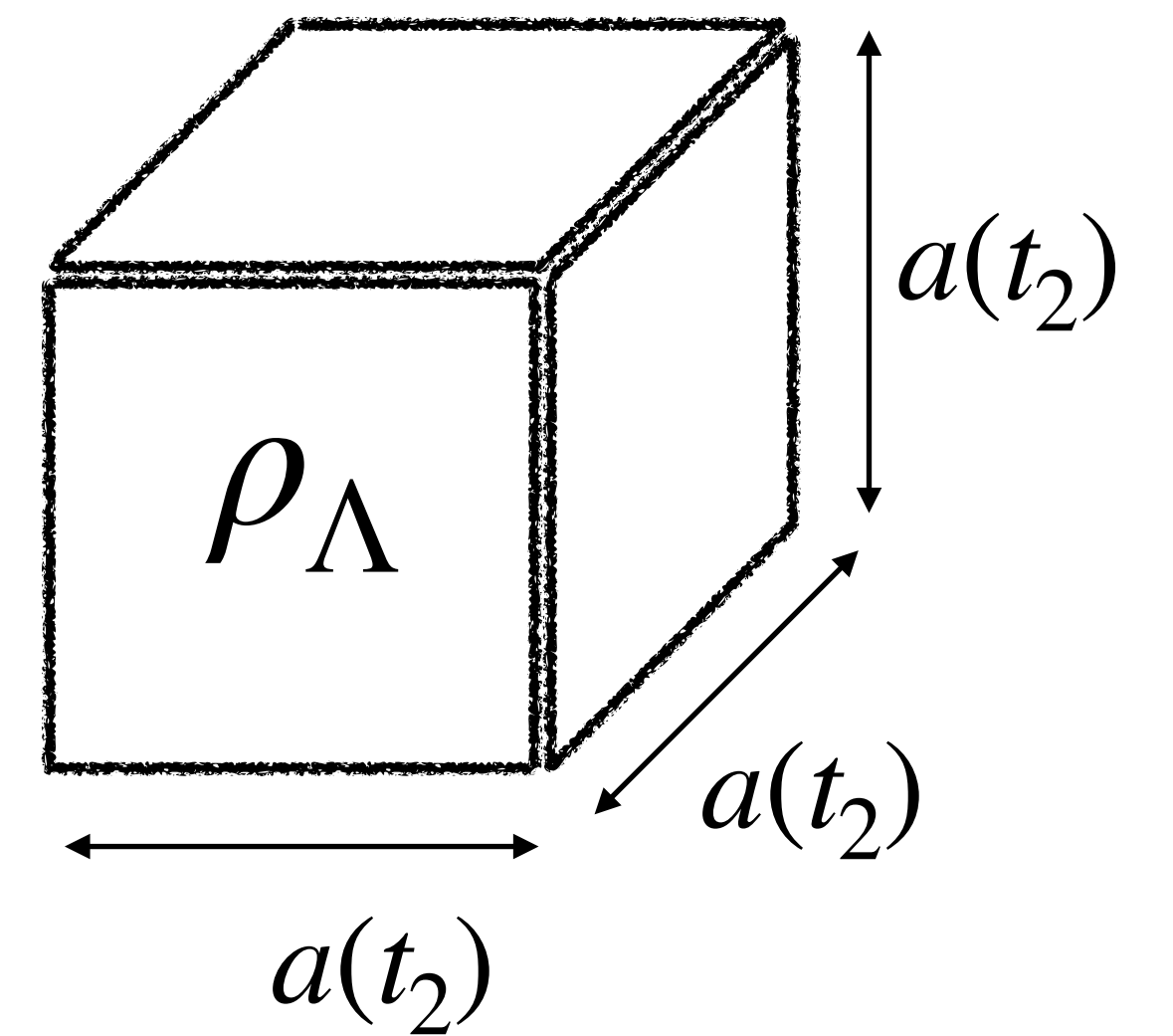
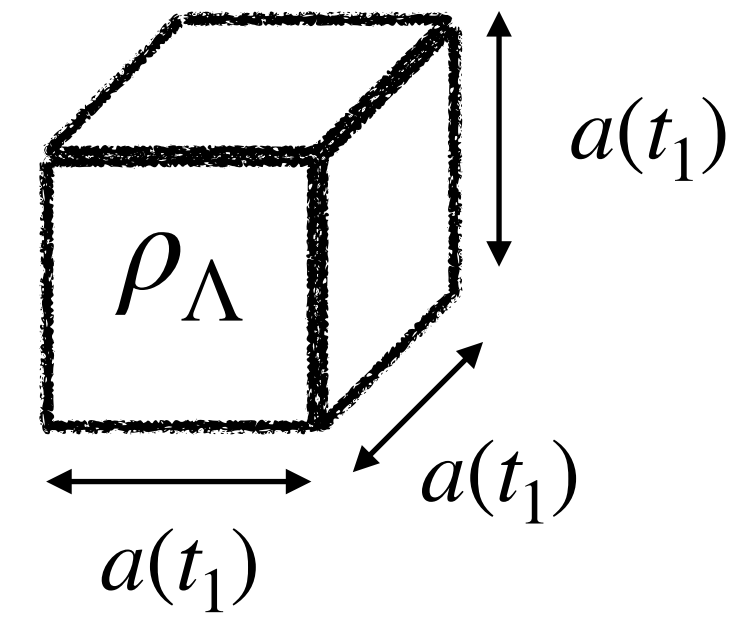
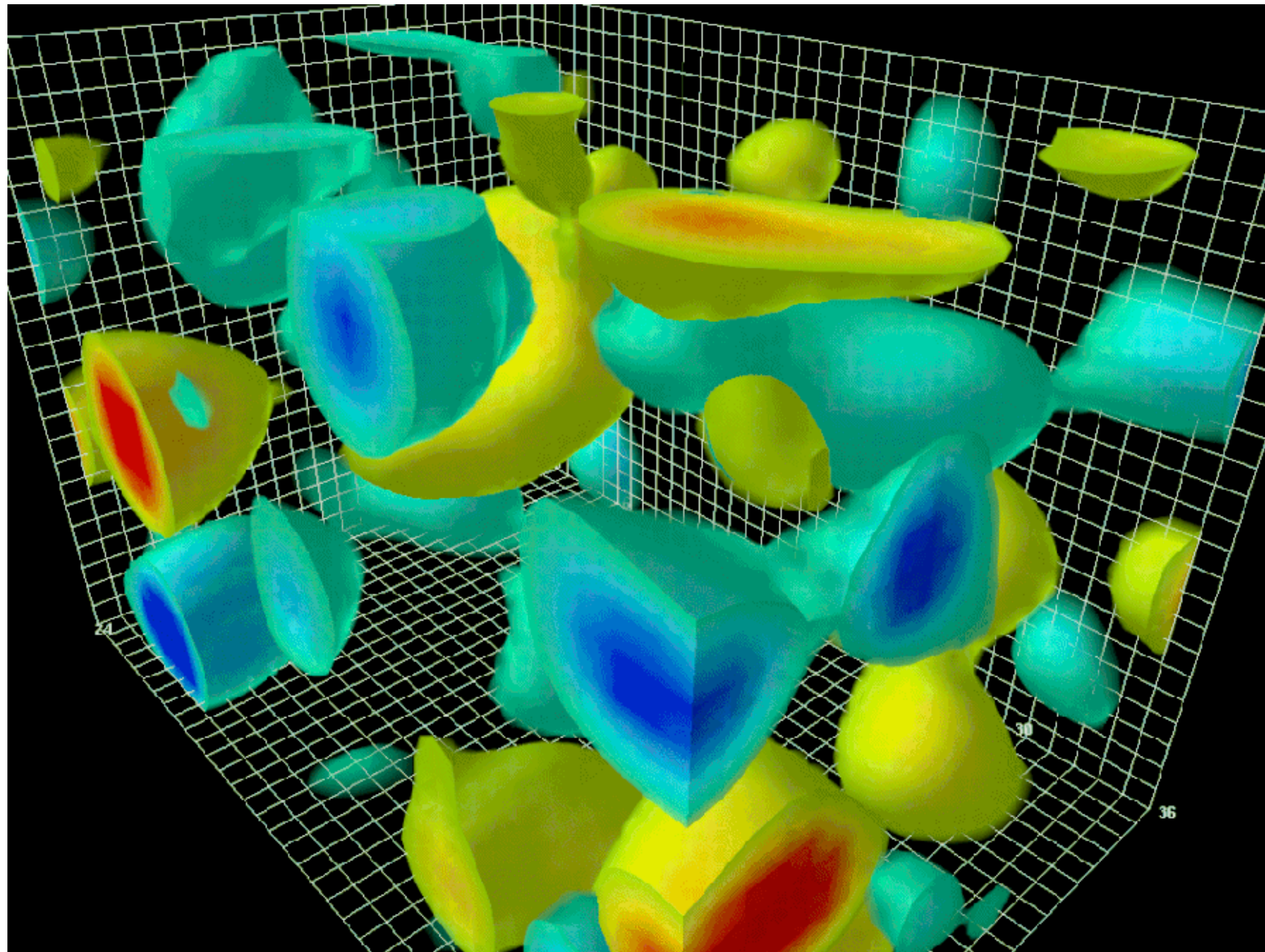
Einstein's equations allow for a constant term

The Universe (and Switzerland) is filled with fields



WHAT IS THE UNIVERSE MADE OF – DARK ENERGY

$$\rho_{\Lambda} \propto a^0$$



TOWARDS COSMOLOGY – FRIEDMANN EQUATIONS

$$\left(\frac{\dot{a}}{a}\right)^2 = \sum_i \frac{8\pi G}{3} \rho_i$$

$$\frac{\ddot{a}}{a} = \sum_i -\frac{4\pi G}{3} (\rho_i + 3p_i)$$

Cold Dark Matter:

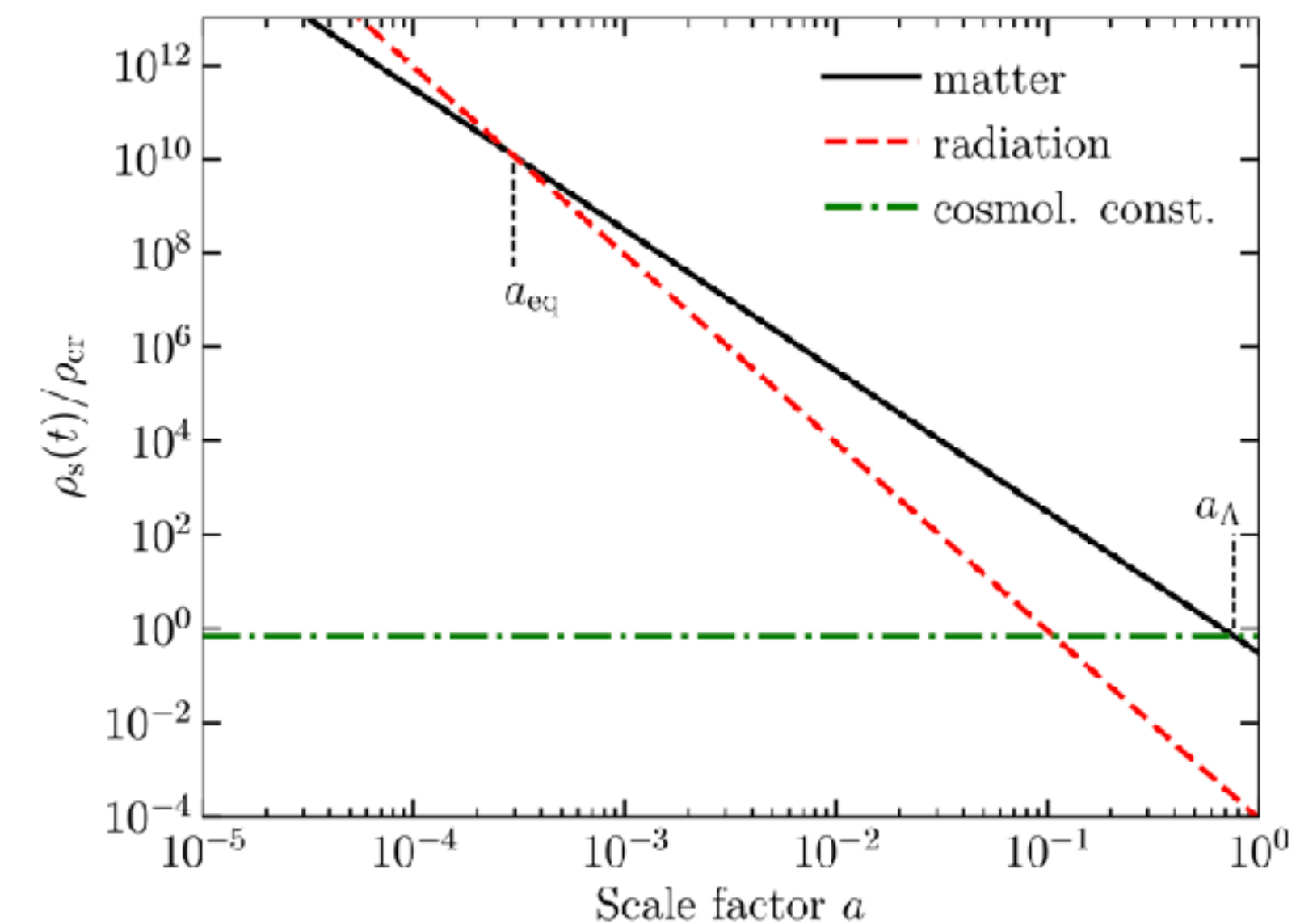
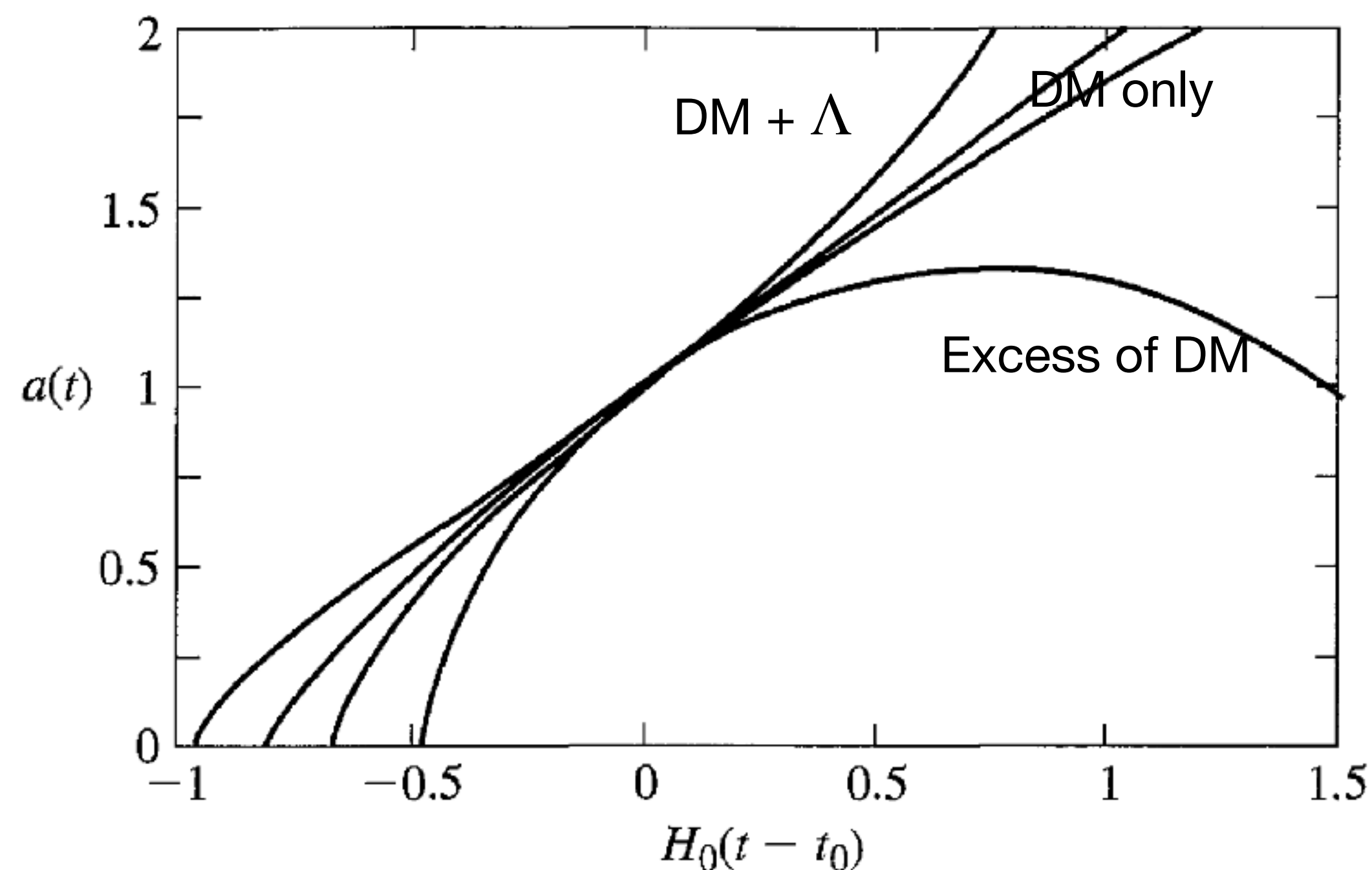
$$\rho_m \propto a^{-3} \quad p_m = 0$$

Dark Energy:

$$\rho_\Lambda \propto \exp(H_0 t) \quad p_\Lambda = -\rho_\Lambda$$

Radiation:

$$\rho_r \propto a^{-4} \quad p_r = -\frac{1}{3}\rho_r$$



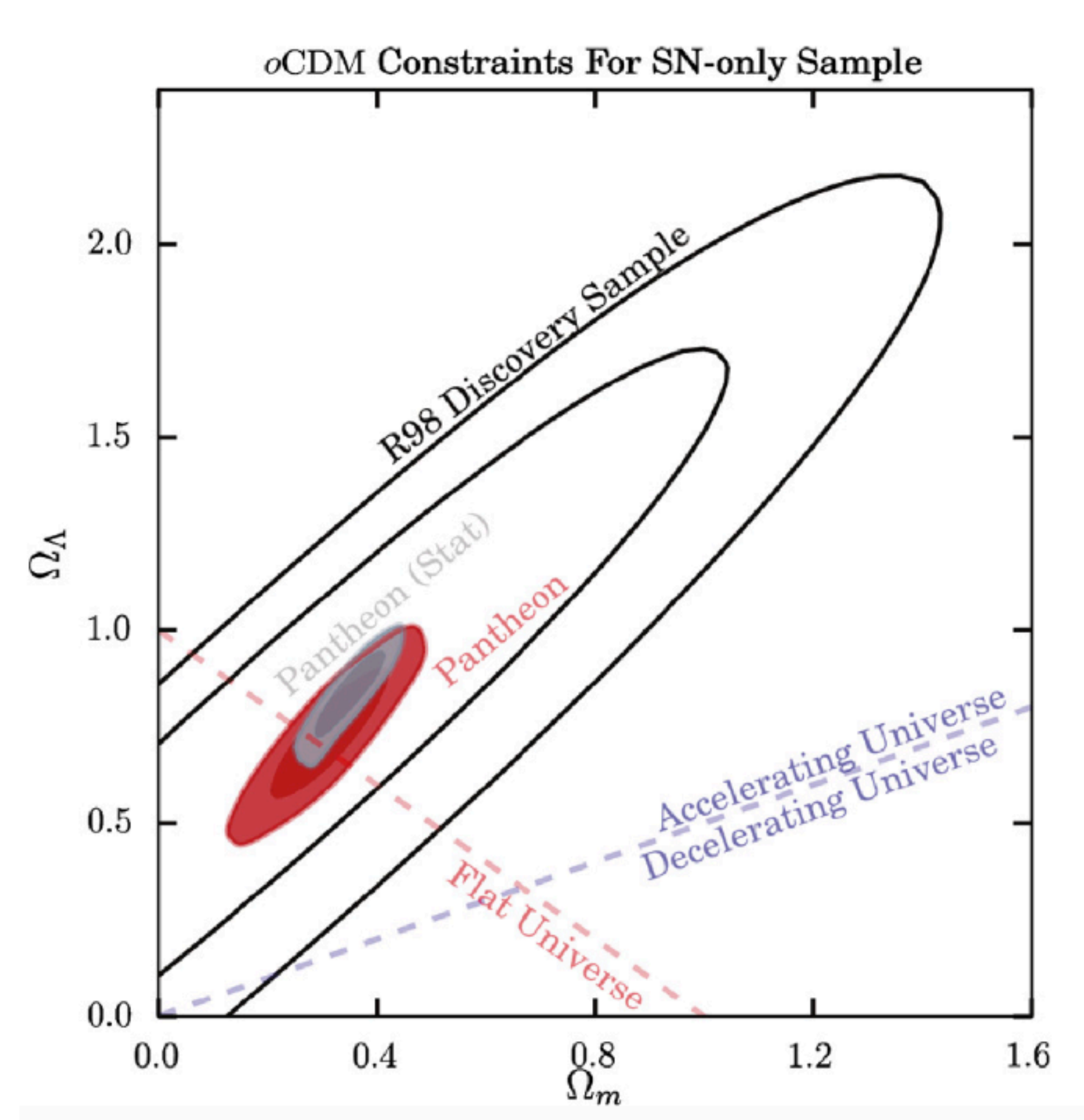
ACCELERATED EXPANSION



Credit: NASA/JPL



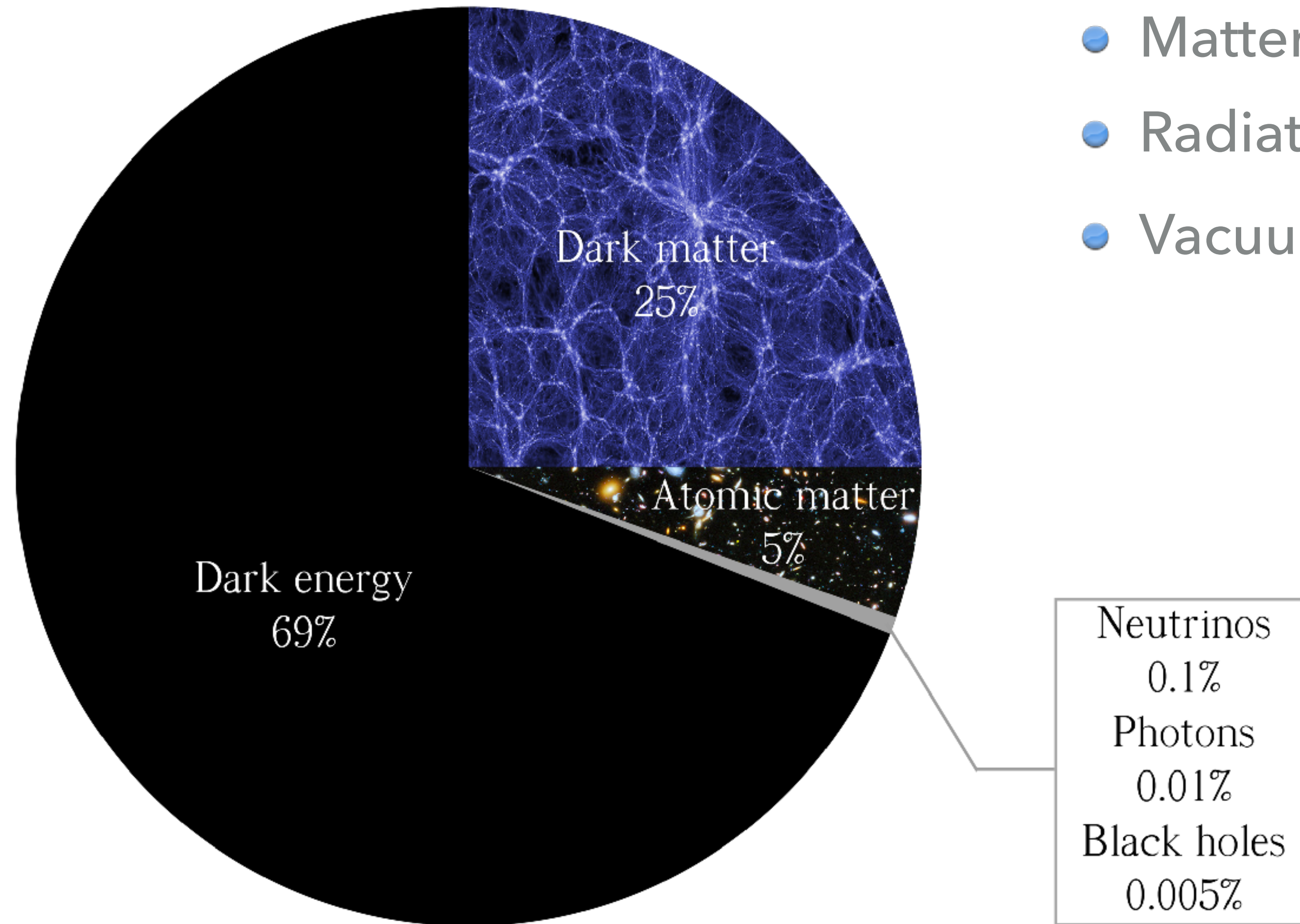
Image by B. J. Fulton, Las Cumbres Observatory Global Telescope Network.



Credit :Dodelson and Schmidt

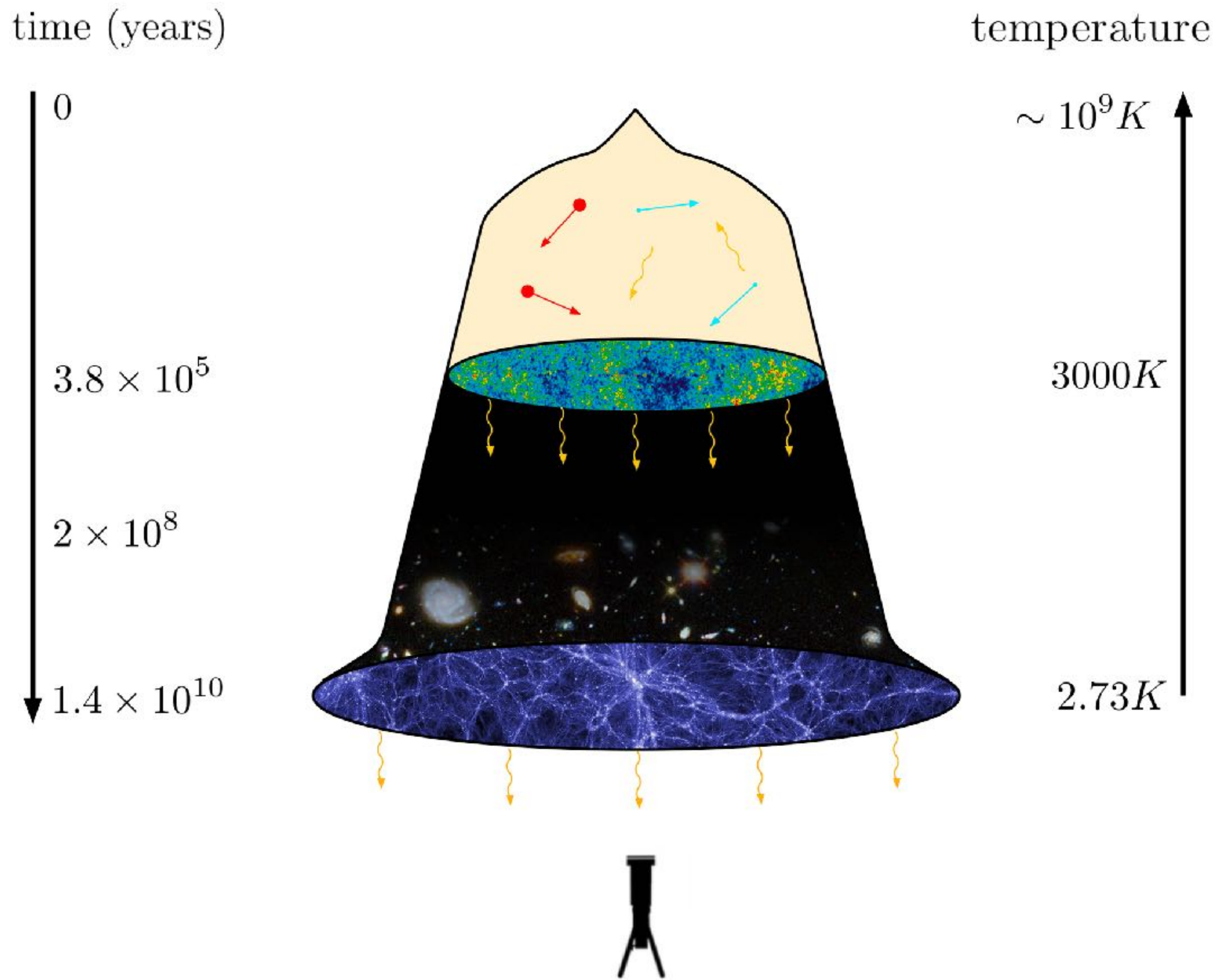
$$m - M = 5 \log \left(\frac{d_L}{10 \text{ pc}} \right) \quad d_L \propto \int_0^z \frac{dz'}{H(z')}$$

WHAT IS THE UNIVERSE MADE OF – MATTER/ENERGY CONTENT

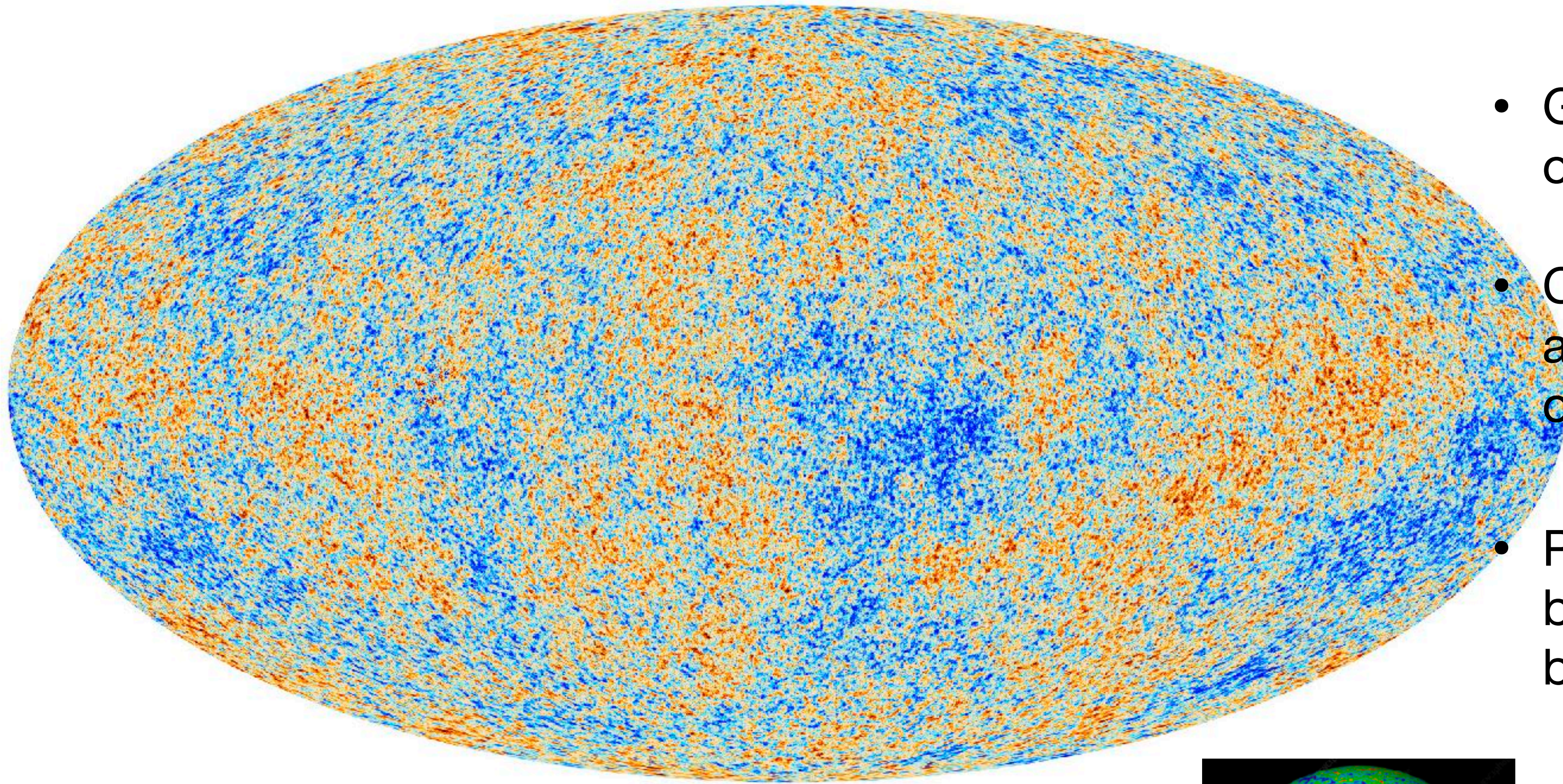


- Matter: moves slowly ($v \ll c=1$)
- Radiation: $v=c=1$ (massless)
- Vacuum (Dark Energy)

THE THERMAL HISTORY OF THE UNIVERSE

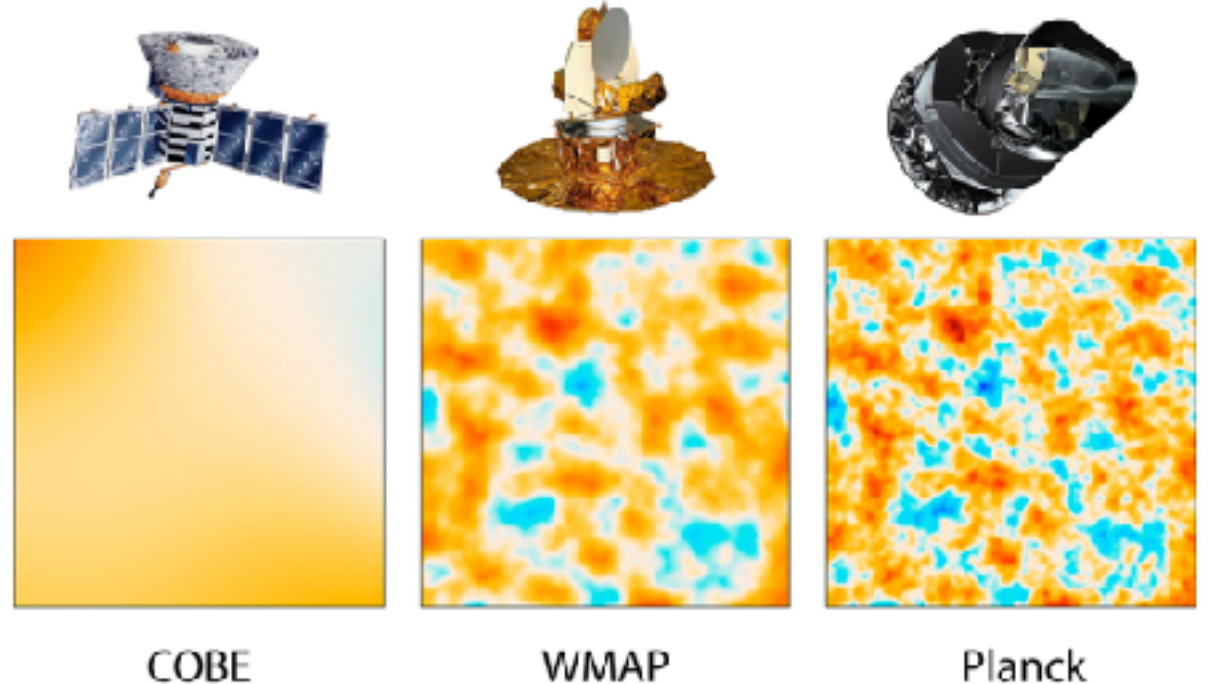
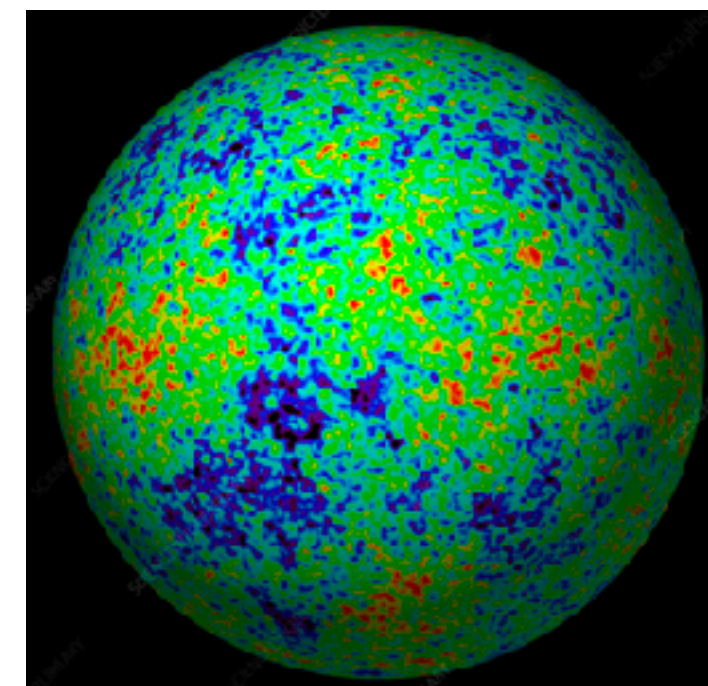


THE COSMIC MICROWAVE BACKGROUND

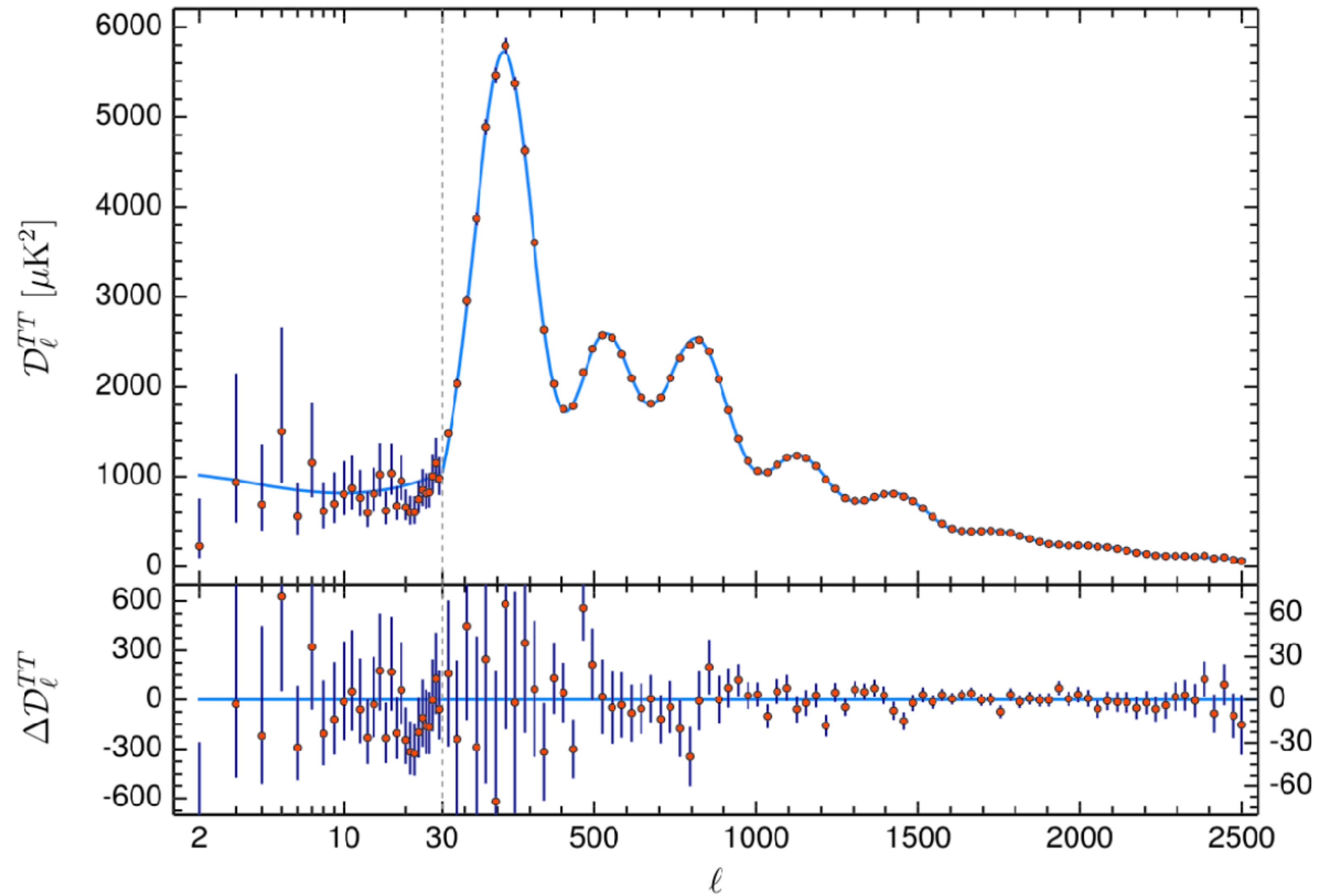


- Gravitational instability: overdense regions contract
- Contracting regions heat up and bounce back (acoustic oscillations)
- Perturbations damp, because radiation flows between them.

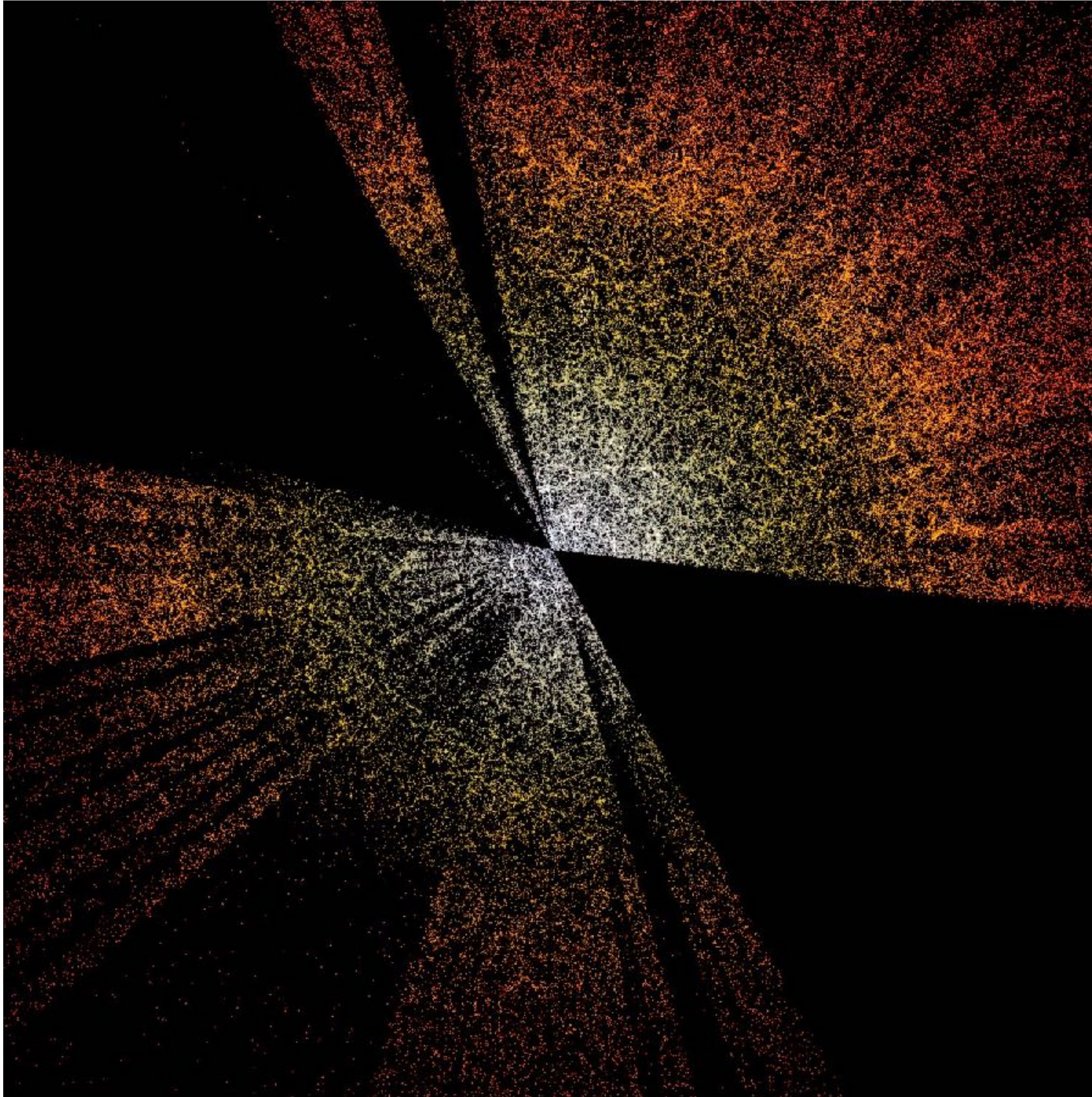
$$\delta(x, t) = \frac{\rho(x, t) - \bar{\rho}(t)}{\bar{\rho}(t)} \quad \delta(t_{rec}) \simeq 10^{-5}$$



THE COSMIC MICROWAVE BACKGROUND



GALAXY SURVEYS

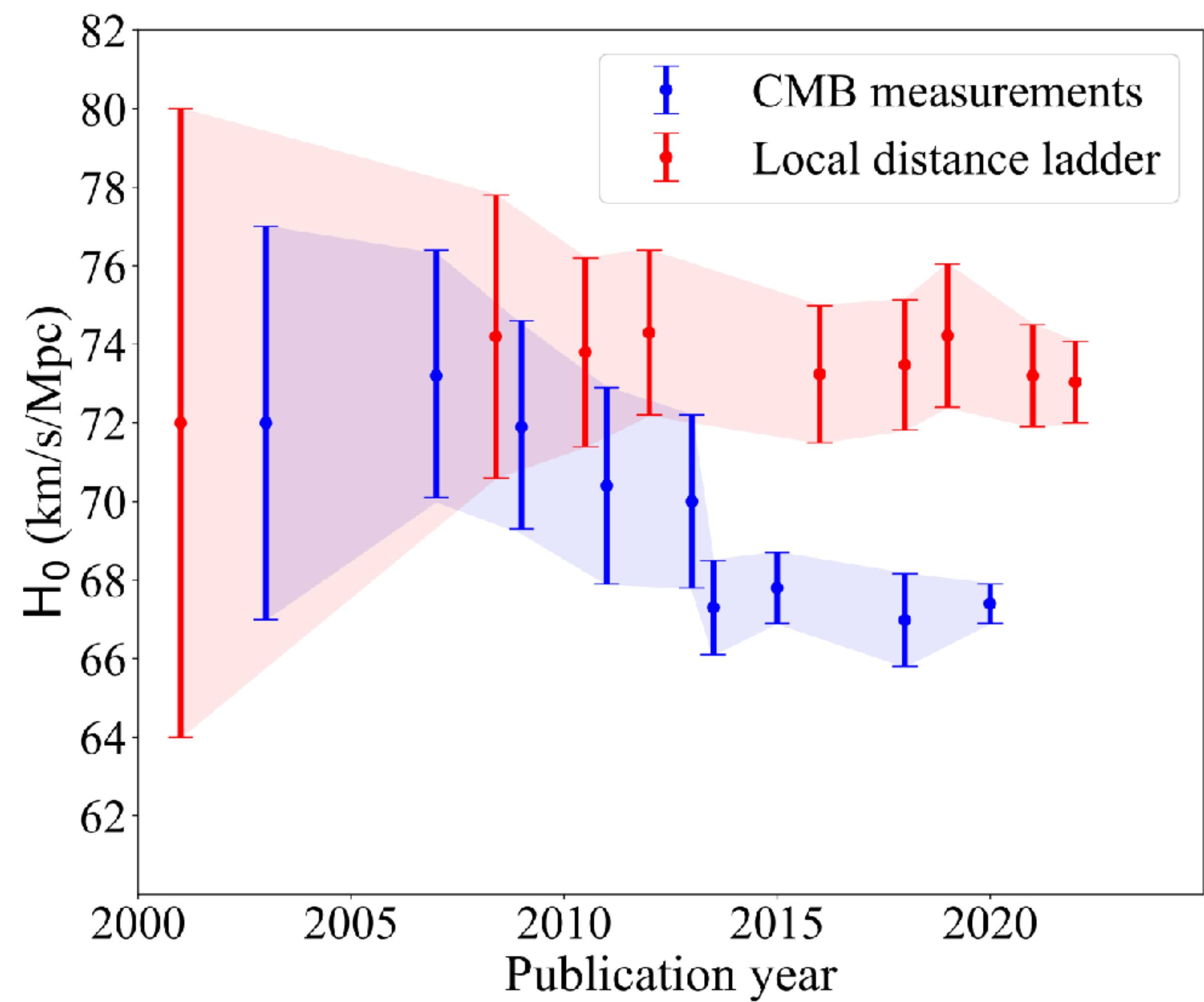


Credit: DESI collaboration

Big questions that we are trying to answer:

- What is Dark Energy? Is it a cosmological constant, or something that changes over time? Or ...
- Is General relativity modified on large-scales?
- Is the cosmological principle accurate on large-scales?
- Is inflation right?
- Is our Λ CDM an accurate description of the Universe?

HINTS THAT Λ CDM MIGHT BE WRONG: H_0 TENSION



Credit: Perivolaropoulos and Skara (2022)

SUMMARY

- General Relativity (GR) is a powerful framework to understand the Universe.
- GR predicts that the Universe is dynamical, either it expands or contracts.
- In 1929 Hubble discovered that the Universe is expanding
- There is plenty of evidence that there is a type of matter that does not interact with electromagnetism (Dark Matter)
- The vacuum energy contributes as a cosmological constant, but the exact cosmological effect of it is poorly understood.
- We can measure the temperature distribution of the cosmic microwave background and compare our theoretical models to it. It gives further evidence of Dark Matter and established Λ CDM as the standard model of cosmology
- We live in the era of the large galaxy surveys and with this data we will have the opportunity to learn more about the nature of dark energy.
- We already have hints that Λ CDM is wrong.